S^{4}D: Style and Structure Similarity for Site Domains

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Περίληψη

Τα τελευταία χρόνια, οι ιστότοποι έχουν γίνει πιο πολύπλοκοι και ποικιλόμορφοι ως προς τον σχεδιασμό και τη δομή τους. Παρά την ποικιλομορφία αυτή, οι ιστότοποι που βρίσκονται στον ίδιο θεματικό τομέα ή εξυπηρετούν παρόμοιους σκοπούς συχνά μοιράζονται κοινά παρουσιαστικά και δομικά μοτίβα. Η αναγνώριση και η σύγκριση αυτών των μοτίβων σε διαφορετικούς τομείς ιστότοπων είναι πολύτιμη για την επίτευξη αποτελεσμάτων ως προς την ταξινόμηση ιστοσελίδων, την αυτοματοποιημένη ανάλυση διεπαφών χρήστη (UIs) και την αποτύπωση των θεματικών τους τομέων. Ωστόσο, η δομή και η οπτική παρουσίαση των περιεχομένων ιστού είναι συχνά μη τυποποιημένη και ασυνεπής, αυξάνοντας τη δυσκολία τέτοιων αναλύσεων. Σε αυτήν την πτυχιακή εργασία, παρουσιάζεται το $\mathbf{S^4D}$, ένα σύστημα που αυτοματοποιεί τη διαδικασία εξαγωγής, μοντελοποίησης και σύγκρισης των παρουσιαστικών και δομικών ομοιοτήτων διαφορετικών ιστοτόπων. Το σύστημα αναλύει ένα σύνολο ιστοσελίδων μέσω τεχνικών αυτοματοποιημένης περιήγησης, μοντελοποιώντας τα CSS χαρακτηριστικά των στοιχείων τους και οργανώνοντάς τα με τρόπο που επιτρέπει την περαιτέρω ανάλυση. Συγκεκριμένα, μετατρέποντας τα στοιχεία σελίδας σε διανυσματικές αναπαραστάσεις και εφαρμόζοντας τεχνικές μη-επιβλεπόμενης συσταδοποίησης, το σύστημα εντοπίζει ουσιαστικές συστάδες των στοιχείων βάσει των οπτικών τους χαρακτηριστικών. Χρησιμοποιώντας αυτήν την συσταδοποιημένη αναπαράσταση, το σύστημα μοντελοποιεί τα παραγόμενα δεδομένα για κάθε ιστότοπο, εξάγει μοτίβα διάταξης στοιχείων και υπολογίζει την ομοιότητα μεταξύ τους χρησιμοποιώντας διαφορετικές μετρικές απόστασης. Τέλος, το σύστημα παρουσιάζει έναν πίνακα ομοιότητας μεταξύ των ιστοτόπων, προσφέροντας πληροφορίες σχετικά με τη δομική και παρουσιαστική ομοιότητα των ιστοσελίδων τους.

Abstract

In recent years, websites have become increasingly complex and diverse in design and structure. Despite this diversity, websites belonging to the same domain or serving similar purposes frequently exhibit similar stylistic and structural characteristics. Identifying and comparing these patterns across different site domains is valuable for tasks such as web classification, automated user interface (UI) analysis, and domain topic labeling. However, the structure and visual presentation of web content is often unstandardized and inconsistent, making such analysis a challenging task. This thesis introduces $\mathbf{S^4D}$, a system that automates the process of extracting, modeling, and comparing the style and structure of websites across different domains. The system parses a set of webpages using browser automation techniques, modeling CSS-related properties of web elements, and structuring them in a way that enables further analysis. By transforming page elements into vector representations and applying unsupervised clustering techniques, the system identifies meaningful groupings of visual and structural components. Using this clustered representation, the system exploits the resulting data modeling of each domain, extracting high-level patterns of element arrangement, and computes the inter-domain similarity using multiple distance metrics. Finally, the system presents a similarity matrix between domains, providing insight into the structural and stylistic resemblance of their webpages.

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Contents

Chapter 1

Introduction

This thesis presents **S^4D**, a system for analyzing and detecting similarities between websites. **S^4D** is a multi-layered system that aims to facilitate the modeling and analysis of websites' visual and structural similarity. Through a variety of functionalities and a Graphical User Interface (GUI), it is able to yield meaningful insight into the websites' visual and structural similarity analysis process, ultimately generating a similarity matrix for all input webpages.

1.1 Problem Statement

The current landscape of web analysis tools focuses primarily on content-based analysis, often overlooking the significance of visual design patterns and structural similarities between websites. In the era of rapidly expanding web content, understanding the structural similarity between websites has become increasingly important for various applications including web mining, content analysis, and automated web processing. However, traditional approaches face several critical challenges.

1.1.1 Web Structure Analysis Challenges

Modern web structure analysis presents several unique technical challenges that our S^4D system specifically addresses:

- **DOM Complexity and Recursive Processing.** Modern websites contain deeply nested DOM structures with thousands of elements, requiring recursive traversal algorithms to accurately capture hierarchical relationships. The sheer depth and breadth of contemporary HTML trees make consistent and scalable structural analysis a non-trivial engineering challenge.

- **CSS Style Diversity and Standardization.** The variety of CSS styling approaches across websites creates inconsistency in structural representation, making cross-site comparisons particularly challenging. Different sites express the same visual properties through differing units, formats, and conventions, requiring unified normalization before any meaningful comparison can take place.

- **Attribute Extraction and Feature Engineering.** Extracting meaningful features from HTML elements involves processing hundreds of CSS properties per element. Identifying which properties carry structural significance — and how to normalize them for comparison — is central to building a reliable similarity model.

- **Scalability and Performance Optimization.** Processing large numbers of websites demands efficient algorithms and robust memory management. As the number of analyzed domains grows, the computational overhead of feature extraction and comparison increases substantially, requiring careful architectural choices to ensure reliable performance at scale.

- **Multi-dimensional Similarity Computation.** Defining meaningful similarity measures for web structures involves multiple algorithmic challenges, since web pages differ across visual, structural, and positional dimensions simultaneously. No single metric fully captures the richness of inter-site similarity, making combined approaches necessary.

- **Clustering Parameter Optimization.** Achieving optimal clustering results requires careful parameter tuning, as different web datasets exhibit varying densities and structural patterns. Selecting appropriate parameters without overfitting to a specific domain is a non-trivial problem that demands principled search strategies.

- **Cross-Domain Pattern Recognition.** Identifying design trends across different domains requires sophisticated comparison frameworks capable of abstracting away domain-specific content while preserving structural and stylistic signals. This becomes especially challenging when the overall design language varies significantly between domains.

1.1.2 Research Motivation

The need for automated web structural analysis stems from several key factors:

- **Content Management**: Organizations need tools to understand and manage large collections of web content based on structural and visual characteristics.

- **Design Pattern Analysis**: Web developers and designers benefit from understanding common structural patterns and visual trends across websites in their domain.

- **Accessibility Evaluation**: Automated structural analysis can help identify accessibility issues and design inconsistencies across multiple web properties.

- **Web Analytics Enhancement**: Structural similarity analysis provides deeper insights into website categorization and user experience patterns beyond traditional content-based metrics.

1.2 Our Methodology and Contributions

S^{4}D addresses the challenges outlined above through a cohesive set of methods and components. To handle DOM complexity, the system uses unique ID generation and hierarchical tree processing to manage parent-child relationships across depth levels 1–10 while maintaining structural integrity throughout analysis. For CSS diversity, sophisticated parsing and normalization pipelines process pixel values (`px`), percentages (`%`), and relative units in a unified manner, enabling consistent cross-site comparisons.

Attribute extraction is guided by selective filtering focused on structurally significant properties such as `font-size`, `text-align`, `line-height`, and `cursor`, with dedicated handling for color space conversions and unit standardization. Scalability is achieved through domain-based processing modes, ChromeDriver optimization, and parallel data processing, enabling reliable batch analysis across large website collections.

For similarity computation, the system combines HDBSCAN clustering with DBCV (Density-Based Cluster Validation) scoring [11], cosine similarity matrices, and role vector analysis to jointly capture structural, visual, and positional relationships between elements. Clustering parameters are tuned via grid search over cluster size ranges and selection epsilon values, with automatic outlier reassignment to the nearest cluster centroid using Euclidean distance. Cross-domain analysis is enabled through domain-level aggregation, transition matrices between design elements, and the use of the cosine similarity metric.

This thesis makes the following key contributions:

- **Novel feature extraction methodology**: Development of comprehensive techniques for extracting both structural and visual features from websites
- **Advanced similarity metrics**: Introduction of new metrics specifically designed for website comparison
- **Integrated clustering approach**: Implementation of clustering algorithms optimized for website analysis
- **Comprehensive evaluation framework**: Establishment of evaluation metrics and datasets for assessing website similarity systems
- **Practical applications**: Demonstration of real-world applications in web design analysis and automated categorization

1.3 Research Objectives

To tackle the challenges raised above, the primary objectives of this research are:

1. To develop a robust system for extracting and analyzing structural and visual features from websites
2. To create effective similarity metrics that can accurately measure relationships between different websites
3. To implement clustering algorithms that can group websites based on their design and structural characteristics
4. To provide a comprehensive comparison framework for website analysis across different domains
5. To validate the effectiveness of the proposed system through extensive testing and evaluation

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Review** presents a comprehensive review of existing approaches to website analysis, similarity measurement, and clustering techniques relevant to web data.
- **Chapter 3: Methodology** describes the proposed S^{4}D system architecture, including feature extraction methods, similarity metrics, and clustering algorithms.
- **Chapter 4: Implementation** details the technical implementation of the system, including data structures, algorithms, and software architecture decisions.
- **Chapter 5: Experimental Design** outlines the experimental methodology, datasets used, and evaluation metrics employed to assess the system's performance.
- **Chapter 6: Results and Analysis** presents the experimental results, performance analysis, and comparison with existing approaches.

- **Chapter 7: Discussion** provides an in-depth discussion of the findings, limitations, and implications of the research.
- **Chapter 8: Conclusion and Future Work** summarizes the key findings, contributions, and suggests directions for future research.

Chapter 2

Related Work

This chapter reviews research related to the structural and visual analysis of web pages, highlighting prior work in web parsing, information extraction, representation learning, DOM-based similarity, and density-based clustering. The discussion emphasizes how these approaches inform, contrast with, and complement the S^4D system. The review is organized into four major areas: (1) structural understanding and representation of web documents, (2) DOM element similarity and tree-based comparison, (3) web template and design pattern detection, and (4) density-based clustering and its validation. A comparative discussion and identification of research gaps close the chapter.

2.1 Structural Understanding and Representation of Web Documents

Understanding the internal structure of web pages has been a central theme in web analysis research, with approaches ranging from dataset construction to deep learning representations and clustering-based organization.

One of the earliest challenges in web analysis has been the ability to combine textual semantics with structural information. Chen et al. [3] introduced **WebSRC**, a large-scale dataset for structural reading comprehension comprising 400K question–answer pairs across 6.4K web pages, combining HTML source code, screenshots, and metadata. Their work demonstrates that effective web understanding requires not only semantic interpretation but also the hierarchical organization of HTML elements. While WebSRC focuses on question answering, its emphasis on DOM-aware structural comprehension is directly relevant to S^4D's approach of exploiting the hierarchy of DOM nodes as a primary signal for layout analysis. Unlike WebSRC, however, S^4D does not address query answering but targets structural and stylistic similarity across entire domains.

Building on this idea of structural representation, Deng et al. [4] proposed **DOM-LM**, a transformer-based model that jointly encodes both textual content and DOM structure to learn generalizable representations of web pages. DOM-LM addresses the shortcomings of methods that rely solely on visual rendering or plain-text representations, achieving strong generalization in few-shot and zero-shot tasks such as attribute extraction and information retrieval. In contrast, S^4D applies structural encoding not for semantic extraction but to identify recurring patterns of element arrangement across domains. Rather than fine-tuned embeddings, S^4D operates on raw CSS feature vectors derived directly from DOM elements, enabling an unsupervised, label-free pipeline.

Hambley et al. [7] approached the problem of structural web representation from a different angle. In their work on **web structure-derived clustering for optimised accessibility evaluation**, they empirically compared five representations—tag sequences, tree structure, full DOM tree encodings, content-based, and combined structure–content—and evaluated their suitability for clustering large web sites. Their central finding is that DOM structural features (tag sequences, tree topology) produce higher-quality and more coherent clusters than content-based alternatives; furthermore, incorporating structural components significantly improves the representativeness of the resulting sample when auditing accessibility. This directly validates S^4D's architectural choice of working at the level of structural CSS features rather than page text. Hambley et al. apply clustering to reduce thousands of pages to archetypal representatives for accessibility audit on sites of up to 500 pages, whereas S^4D scales clustering to the element level and aggregates results across domains.

Huynh et al. [8] introduced a DOM-structural cohesion analysis approach to **web page segmentation**, treating element types, attributes, and structural cohesion as primary features. Their method achieves 64% F_{B^3} on a benchmark of 1,969 web pages, surpassing prior state-of-the-art segmentation methods, without relying on screenshot-based rendering. Their work confirms that element-level DOM attributes are strong enough signals to segment and partition page structure in a purely structural pipeline—the same philosophy underlying S^4D's CSS feature extraction stage. The key difference is scope: Huynh et al. produce intra-page segmentation boundaries, whereas S^4D uses element-level features for cross-domain clustering and similarity comparison.

2.2 DOM Element Similarity and Tree-Based Comparison

Measuring how similar DOM elements or entire DOM trees are—without relying on rendered images—has attracted a growing body of work.

Grigera et al. [5] proposed two algorithms for the **flexible detection of similar DOM elements**, measuring similarity through a mixed approach that combines an element's location in the DOM tree with its inner structure. The approach does not require CSS attributes or visual screenshots and was evaluated against a manually annotated ground truth of over 1,200 DOM elements, yielding measurable precision across heterogeneous real-world sites. Their work demonstrates that DOM tree position and sub-tree topology already carry enough discriminative power to identify structurally equivalent elements across pages. S^4D extends this idea by adding the CSS attribute dimension—treating CSS values as a feature vector alongside structural position—which enables finer-grained distinctions between elements that share topology but differ in visual style.

Brisset et al. [1] proposed **SFTM** (Similarity-based Flexible Tree Matching), a fast algorithm for matching entire web page DOM trees based on node labels and local topology similarity rather than CSS attributes or visual features. SFTM avoids the combinatorial explosion of exhaustive subtree comparison and achieves an average matching time of approximately 182 ms on real-world web documents. The work is notable for demonstrating that practical tree matching at the scale of real web pages is feasible within a structural-only framework. Where SFTM operates at the full-page tree level to determine a single similarity score between two pages, S^4D operates at the element level, clustering individual CSS-annotated nodes and then aggregating cluster memberships into domain-level role vectors for cross-site comparison.

The extraction of structured information from web pages has been another central research theme, with work from Gulhane et al. [6] on **Vertex**, a wrapper induction system deployed at Yahoo! for large-scale record extraction. Vertex introduced algorithms for grouping similar pages, generating XPath-based extraction rules, and adapting to structural changes—extracting over 250 million records from more than 200 websites. While Vertex demonstrates the scalability of web information extraction and implicitly relies on structural regularity to identify common DOM patterns, it is tailored to structured record retrieval. S^4D does not generate wrappers or extract database-like content; it focuses instead on comparing visual and structural layout signatures across entire domains, making these two systems complementary in scope.

2.3 Web Template and Design Pattern Detection

Detecting recurring design patterns and templates across web sites represents a problem closely related to the cross-domain similarity goal of S^4D.

Larroque et al. [9] proposed a method for **web template extraction based on hyperlink graph analysis**. Their approach identifies a small representative set of pages sharing the same template by analyzing the hyperlink structure of a web site, without requiring visual or NLP-based comparison. Template detection is positioned as a prerequisite for downstream tasks including content extraction, block detection, and web page indexing. By exploiting structural redundancy at the graph level, the method can identify common page frames across large crawls. S^4D addresses a related but orthogonal goal: rather than grouping pages within a site that share a template, it compares the structural and stylistic signatures of elements across different domains to assess cross-domain similarity.

More recently, Lockard et al. [10] proposed **ZeroShotCeres**, which tackles the challenge of zero-shot relation extraction from semi-structured web pages. Their model employs graph neural networks to capture both textual semantics and visual cues such as layout, font size, and color, enabling generalization to previously unseen templates without template-specific training data. ZeroShotCeres highlights the importance of structural and visual cues for generalizable web analysis. While their focus is on entity-level relation extraction, the reliance on layout and visual features as cross-template signals parallels S^4D's use of CSS attributes as cross-domain structural fingerprints. S^4D, however, does not require any labeled training data and does not target relation extraction, operating instead in a fully unsupervised regime for domain-level similarity.

2.4 Density-Based Clustering and Clustering Validation

The clustering layer of S^4D relies directly on HDBSCAN and the DBCV validity index. Both have been formally introduced and analyzed in the literature.

Campello et al. [2] introduced **HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise), a density-based clustering algorithm that constructs a complete hierarchy of clusters across all possible density thresholds following Hartigan's density-contour model. From this hierarchy, HDBSCAN extracts a flat partitioning in a fully unsupervised manner by maximizing cluster *stability*—selecting clusters that persist over a wider density range (i.e., whose stability exceeds the sum of their children's stabilities)—without requiring the user to specify an epsilon threshold as in traditional DBSCAN. In addition, HDBSCAN assigns each point a GLOSH outlier score in $[0, 1]$ that unifies global and local outlier perspectives. These properties make HDBSCAN well-suited to high-dimensional, heterogeneous web data where clusters vary widely in density and shape, precisely the setting of S^4D's CSS feature space.

Moulavi et al. [11] introduced **DBCV** (Density-Based Clustering Validation), a clustering validity index designed specifically for density-based algorithms. The motivation behind DBCV is that traditional validity indices—including Silhouette, Davies-Bouldin, and Calinski-Harabasz—assume globular or spherical cluster geometry and are therefore inappropriate for evaluating the output of algorithms like HDBSCAN, which can discover arbitrarily shaped clusters. DBCV provides a principled alternative that respects the density-based structure of the solution. S^4D adopts DBCV as its cluster quality metric to guide hyperparameter selection (notably the minimum cluster size) and to compare alternative

CSS feature configurations, ensuring that the chosen clustering reflects genuine density structure in the element-level feature space rather than artifacts of spherical assumptions.

2.5 Comparison with Prior Work

The reviewed works collectively advance web structural analysis along multiple complementary dimensions, yet none directly targets the specific combination of concerns addressed by S^{4}D:

- **WebSRC** [3]: Focuses on question answering over structured web data, emphasizing the need for DOM-aware structural comprehension. S^{4}D uses structural encoding not for query answering but for unsupervised cross-domain similarity analysis.

- **DOM-LM** [4]: Learns generalizable document embeddings from DOM and text jointly, requiring fine-tuned transformer training. S^{4}D operates on raw CSS feature vectors with no training phase, applying unsupervised clustering directly.

- **Hambley et al.** [7]: Demonstrate empirically that DOM structural features outperform content features for page clustering, applied to accessibility evaluation within single sites. S^{4}D extends this structural clustering paradigm to the element level and aggregates results for cross-domain comparison.

- **Huynh et al.** [8]: Perform intra-page segmentation via DOM-structural cohesion analysis, using element types and attributes as features. S^{4}D shares the element-level feature perspective but applies it to inter-domain clustering rather than intra-page boundary detection.

- **Grigera et al.** [5]: Measure DOM element similarity via tree position and inner structure. S^{4}D augments this structural signal with CSS attribute values, enabling style-sensitive similarity alongside topological similarity.

- **Brisset et al. (SFTM)** [1]: Match entire page DOM trees efficiently using node labels and local topology. S^{4}D operates at the element level rather than the full-page tree level, clustering fine-grained CSS-annotated nodes before aggregating to domain-level role vectors.

- **Vertex** [6]: Extracts structured records at web scale via wrapper induction. S^{4}D does not extract records but compares layout and style fingerprints across domains.

- **ZeroShotCeres** [10]: Performs zero-shot relation extraction using layout and visual features modeled by GNNs. S^{4}D uses similar structural cues but requires no labeled data and does not target entity-level relations.

- **Larroque et al.** [9]: Detects intra-site template groups via hyperlink graph analysis. S^{4}D targets cross-domain similarity rather than within-site template clustering.

- **HDBSCAN** [2]: The foundational density-based clustering algorithm adopted by S^{4}D. The paper establishes the stability-maximization extraction strategy that S^{4}D directly exploits for element-level CSS clustering.

- **DBCV** [11]: The validity index used by S^{4}D to evaluate and tune its HDBSCAN configurations, selected because it handles the non-globular cluster shapes that arise in CSS feature spaces.

2.6 Research Gaps and Opportunities

The literature reveals several gaps that motivate the design of S^{4}D:

- **Cross-domain structural similarity**: The majority of reviewed systems operate within a single website or across pages sharing a known template. None compute a similarity matrix across structurally heterogeneous domains, which is the primary output of S^{4}D.
- **Element-level CSS feature clustering**: Existing work on DOM element similarity (Grigera et al., Huynh et al.) relies on tag types and tree topology. Incorporating the CSS attribute dimension—font sizes, margins, padding, display properties—as a continuous feature vector for clustering is largely unexplored.
- **Multi-dimensional aggregation**: Prior clustering-based web analysis (Hambley et al.) clusters entire pages. S^{4}D introduces a two-level approach: clustering at the element level, then aggregating cluster membership distributions into domain-level role vectors for inter-domain comparison.
- **Unsupervised, label-free pipeline**: Systems like DOM-LM and ZeroShotCeres require either labeled data or fine-tuning. S^{4}D achieves cross-domain similarity estimation in a fully unsupervised manner using only raw CSS attributes and the HDBSCAN/DBCV stack.
- **Scalable cross-domain analysis**: While Vertex demonstrates extraction scale, scalable structural similarity across diverse domains remains underexplored. S^{4}D's pipeline is designed to process multiple domains concurrently and aggregate results at the domain level.

2.7 Summary

This chapter has reviewed eleven representative systems and methods that collectively span the major technical dimensions underlying S^{4}D: structural web understanding (WebSRC, DOM-LM, Hambley et al., Huynh et al.), DOM element and tree-based similarity (Grigera et al., Brisset et al./SFTM, Vertex), web template and pattern detection (Larroque et al., ZeroShotCeres), and density-based clustering with validation (HDBSCAN by Campello et al., DBCV by Moulavi et al.). Each system contributes important ideas relevant to S^{4}D, yet none directly combines element-level CSS feature extraction, density-based clustering with DBCV-guided tuning, and cross-domain role-vector aggregation into a single unsupervised pipeline. The gaps identified above motivate the design choices presented in the following chapter.

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